DRS - Digestion and Absorption of Nutrients II

Fats

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Dr Majda Bakali



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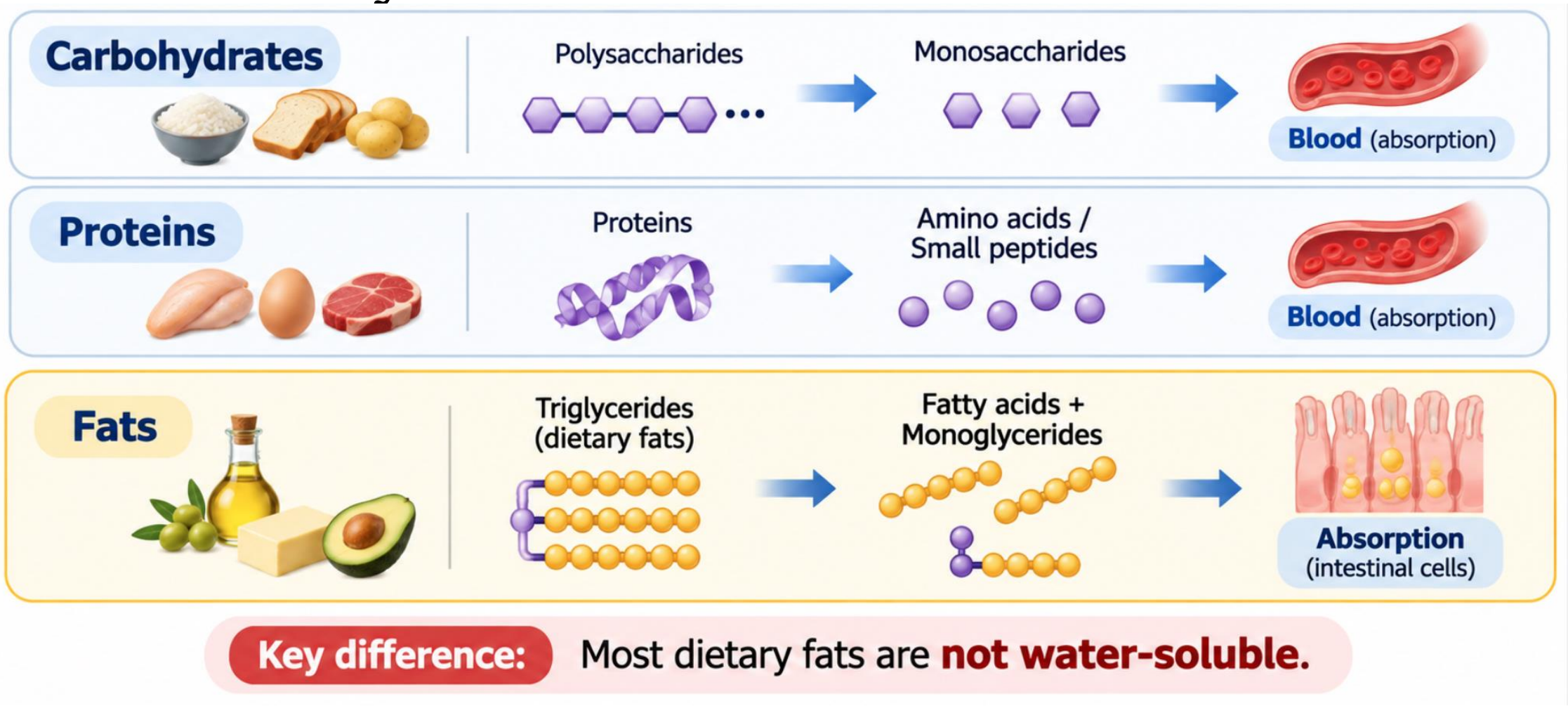
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Learning Outcomes

- Using suitable examples, describe the functions of lipids
- Describe the mechanism of digestion for fats
- Compare absorption of long and short-chain fatty acids
- Discuss the clinical relevance of disorders related to the digestion and absorption of fats
- Explain how vitamin deficiency affects human health

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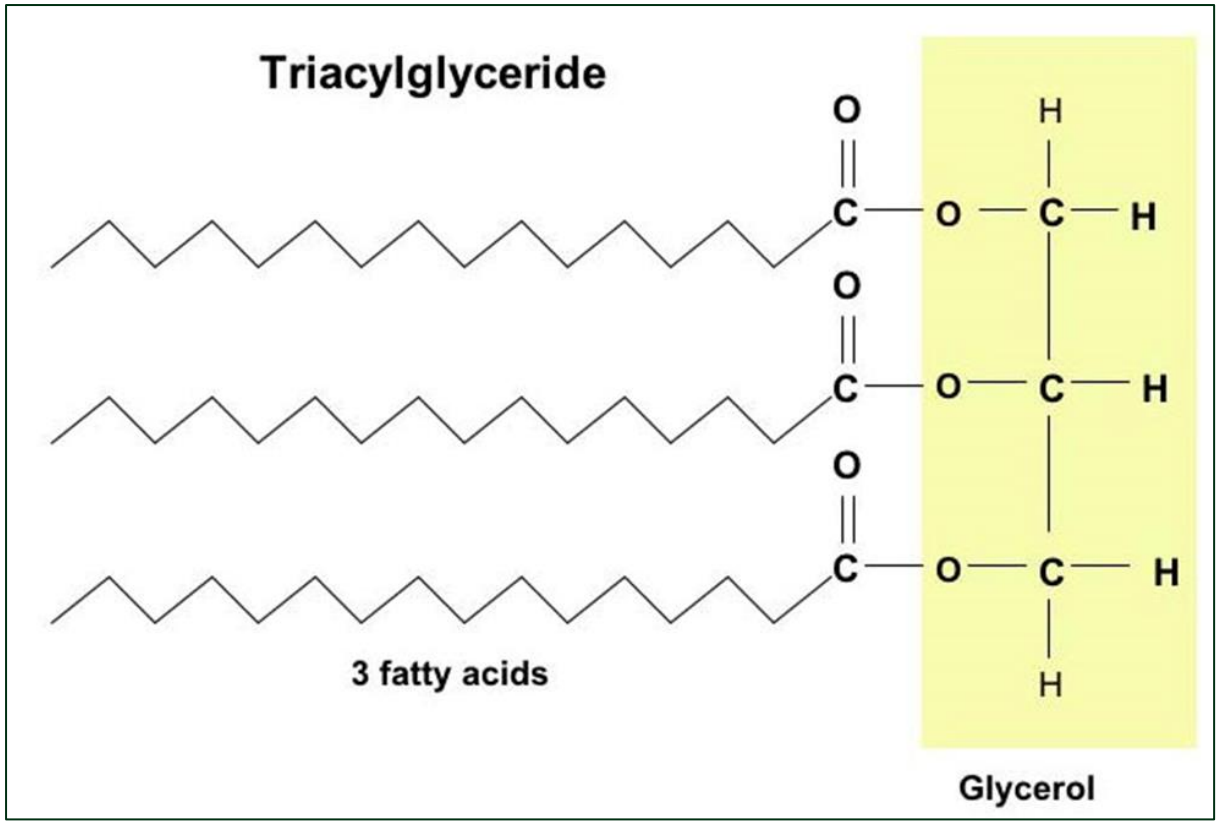
From Carbohydrates and Proteins to Fats



Dietary Fats: Triglycerides

Triglycerides

- Approximately 90% of dietary fat consists of triglycerides.
- Triglyceride = Glycerol + 3 Fatty acids
- Fatty acids differ in chain length and saturation.



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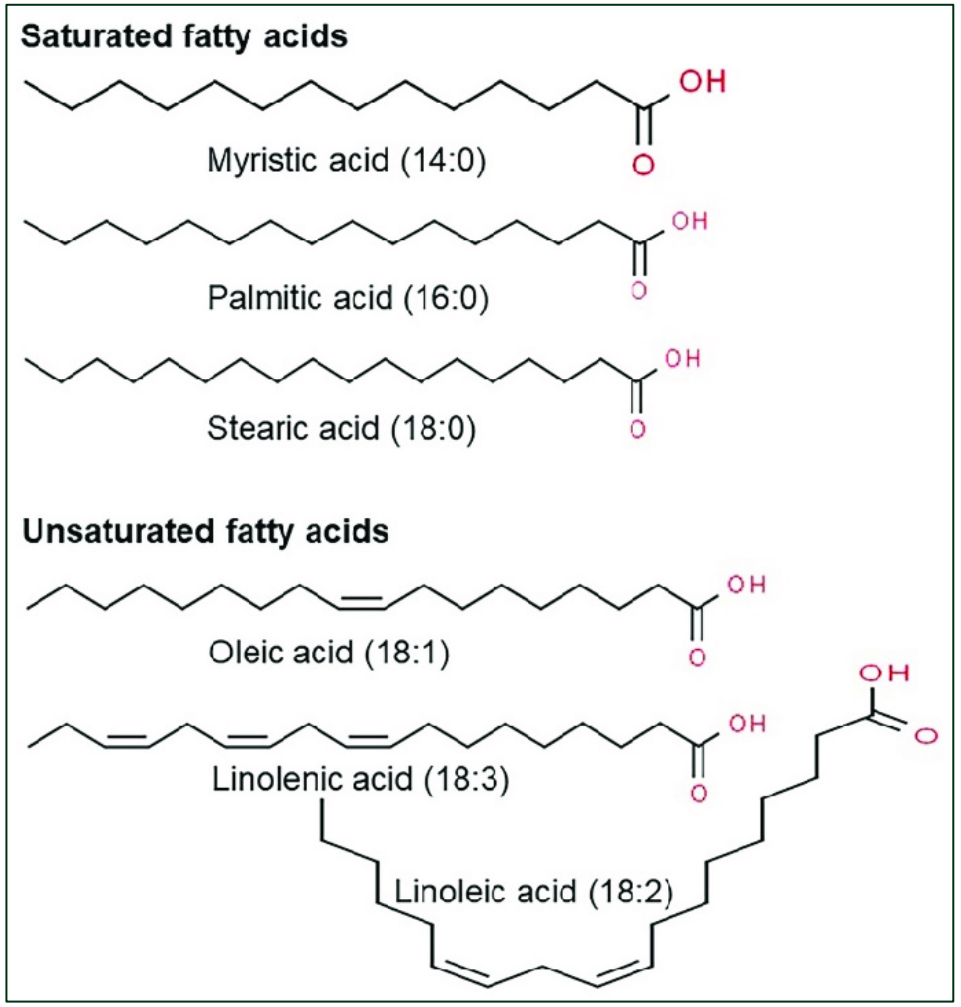
Saturated and Unsaturated Fatty Acids

Saturated fatty acids

- No carbon–carbon double bonds.
- Examples: Palmitic acid (16:0), Stearic acid (18:0).

Unsaturated fatty acids

- Contain one or more carbon–carbon double bonds.
- Examples: Oleic acid (18:1), Linoleic acid (18:2), Linolenic acid (18:3).



Chain length | Double bonds | Saturation

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Essential Fatty Acids

Essential Fatty Acids

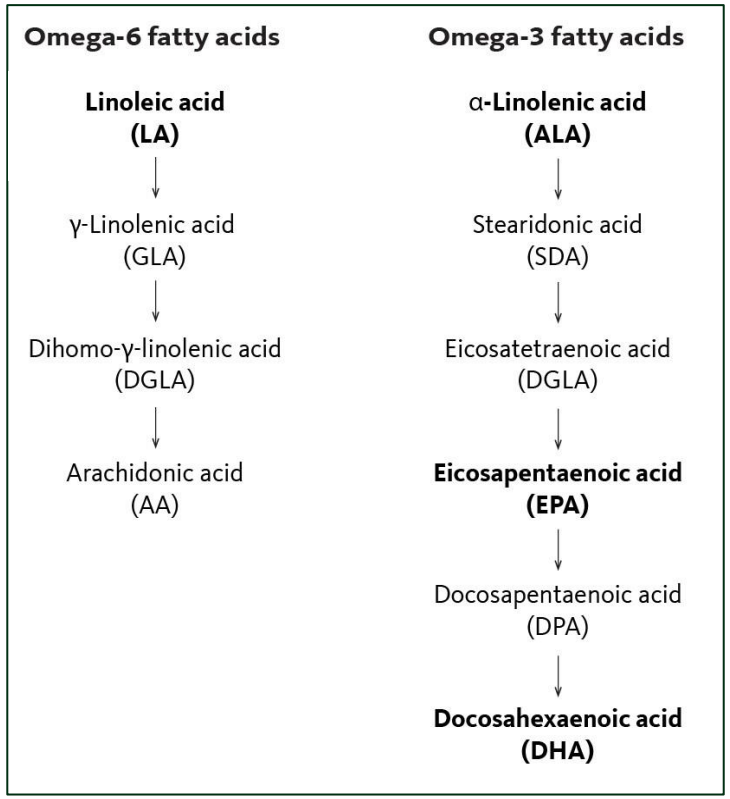
- Cannot be synthesised in sufficient amounts by the human body.
- Must be obtained from the diet.

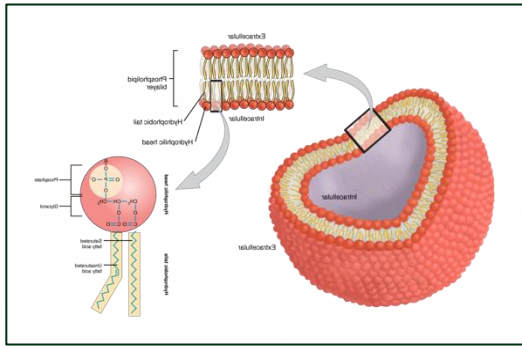
Examples

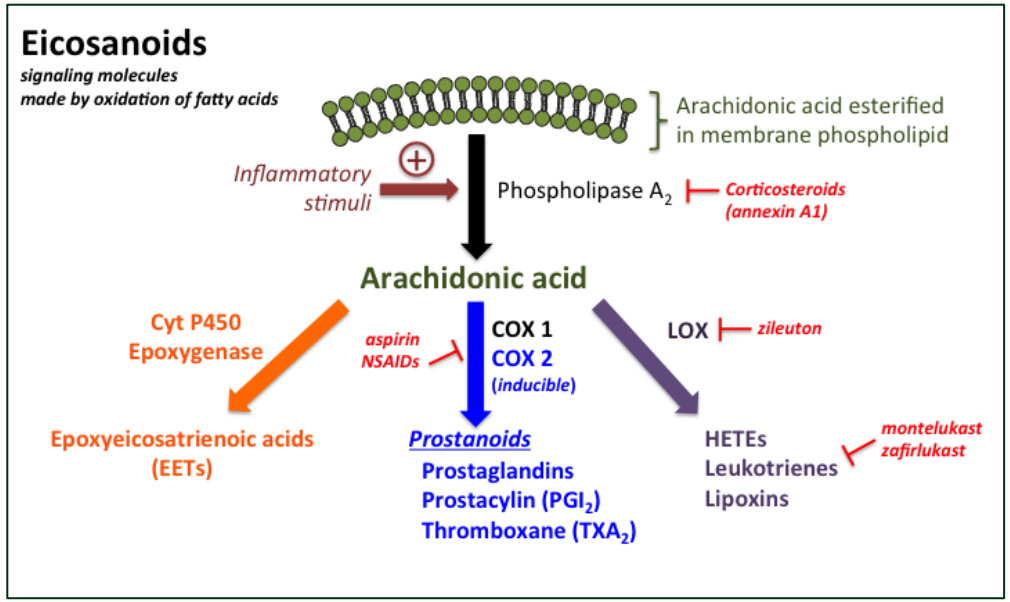
- Linoleic acid (18:2, ω-6)
- α-Linolenic acid (18:3, ω-3)
- Arachidonic acid may become conditionally essential.

Functions

- Cell membrane structure
- Eicosanoid production
- Regulation of blood pressure and blood properties
- Support of erythropoiesis







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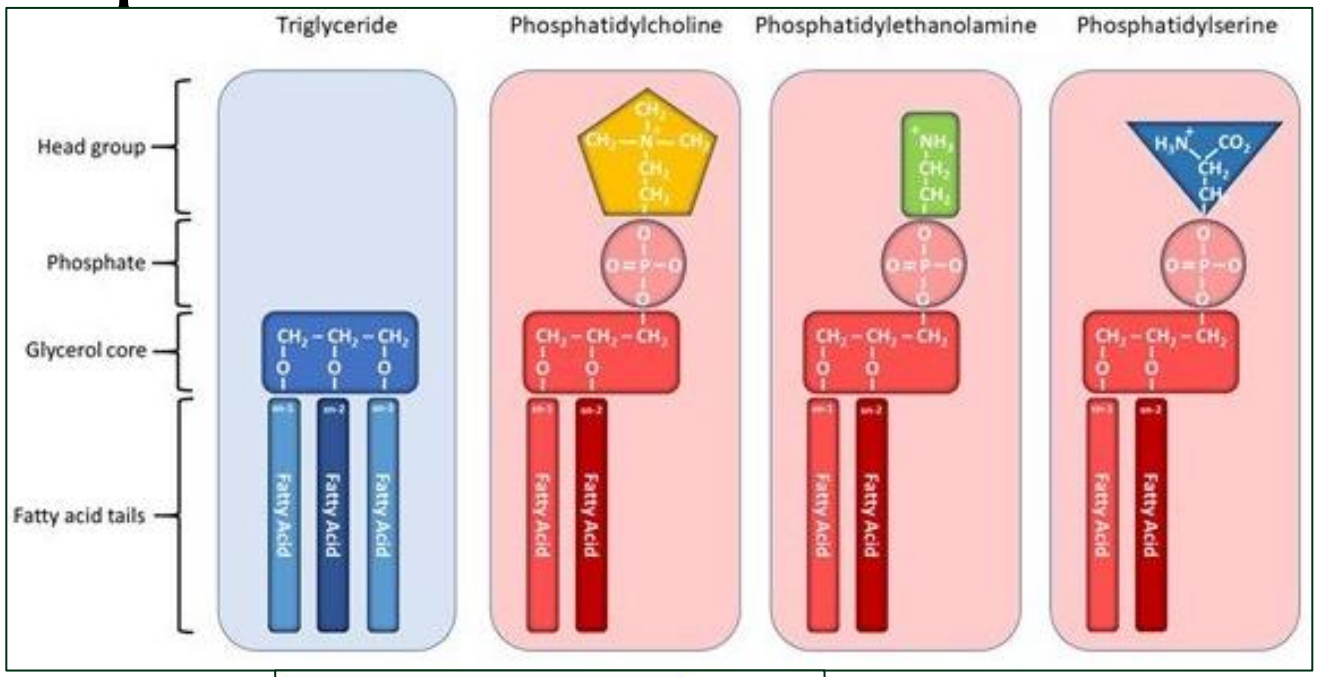
Phospholipids and Sphingolipids

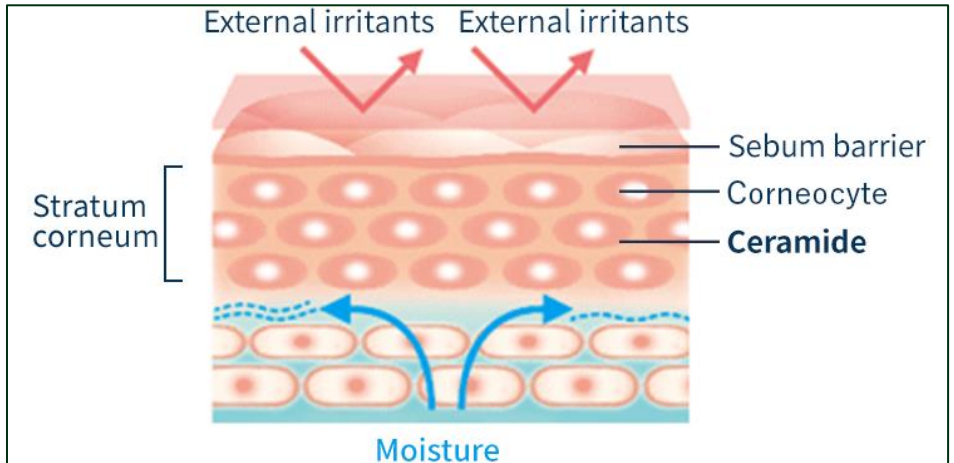
Phospholipids

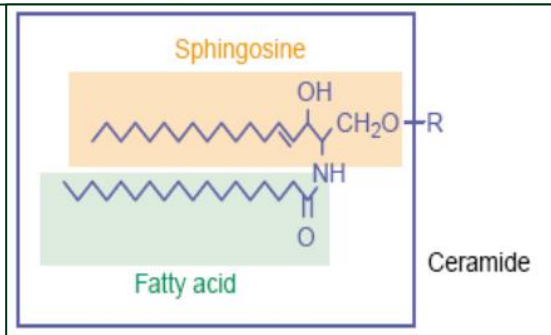
- Approximately 5% of dietary lipids come from cell membranes.
- Two fatty acid tails and a phosphate-containing head group.
- Examples: PC, PE, PS.

Sphingolipids

- Important membrane lipids.
- Have a sphingoid backbone rather than glycerol.
- Example: Ceramide, important in the skin barrier.







Substituent (R)	Sphingolipid
H	Ceramide
Phosphocholine	Sphingomyelin
Sugar(s)	Glycosphingolipid

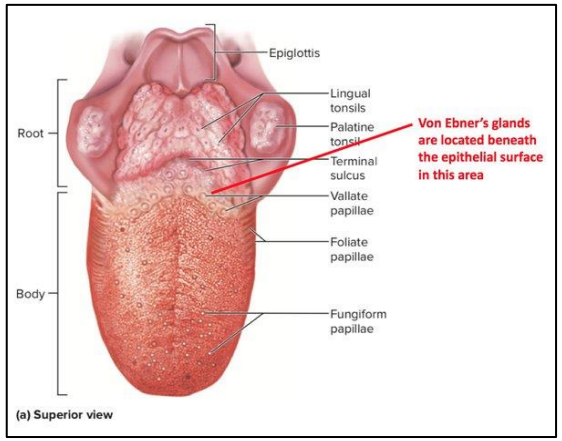
General sphingolipid structure

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Fat Digestion

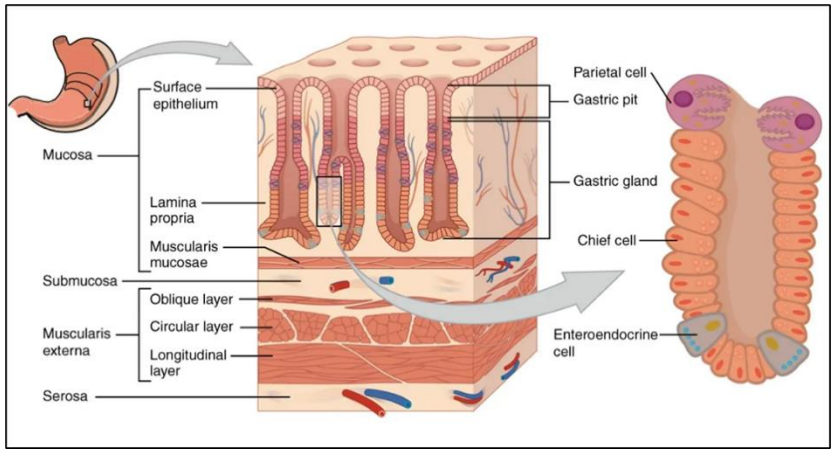
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Three Lipases Involved in Fat Digestion



1. Lingual lipase

- Secreted by glands associated with the tongue.
- Begins fat digestion; active at low pH.

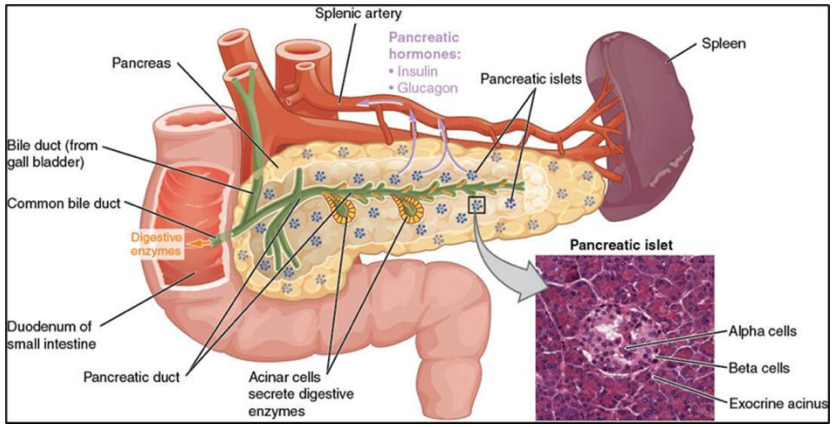


2. Gastric lipase

- Secreted by gastric chief cells.
- Continues fat digestion in the stomach.

3. Pancreatic lipase

- Secreted by pancreatic acinar cells.
- Major enzyme for fat digestion in the small intestine.



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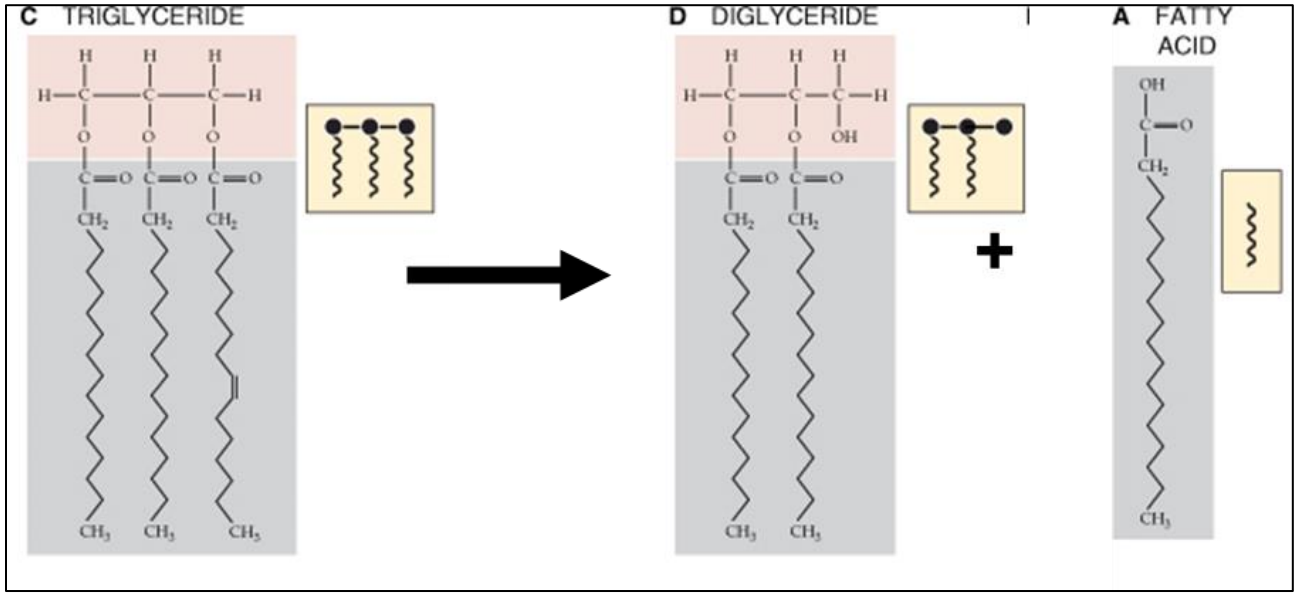
Fat Digestion in the Mouth and Stomach

Early Fat Digestion

- Fat digestion begins in the mouth and continues in the stomach.
- Lingual and gastric lipases hydrolyse triglycerides.

Triglyceride → Diglyceride + Fatty acid

- Long-chain fatty acids remain associated with fat droplets.
- Shorter-chain fatty acids are more soluble and may enter portal blood.
- Approximately 15% of fat digestion occurs before the small intestine.



Short-chain FA → Portal blood

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CCK Coordinates Fat Digestion

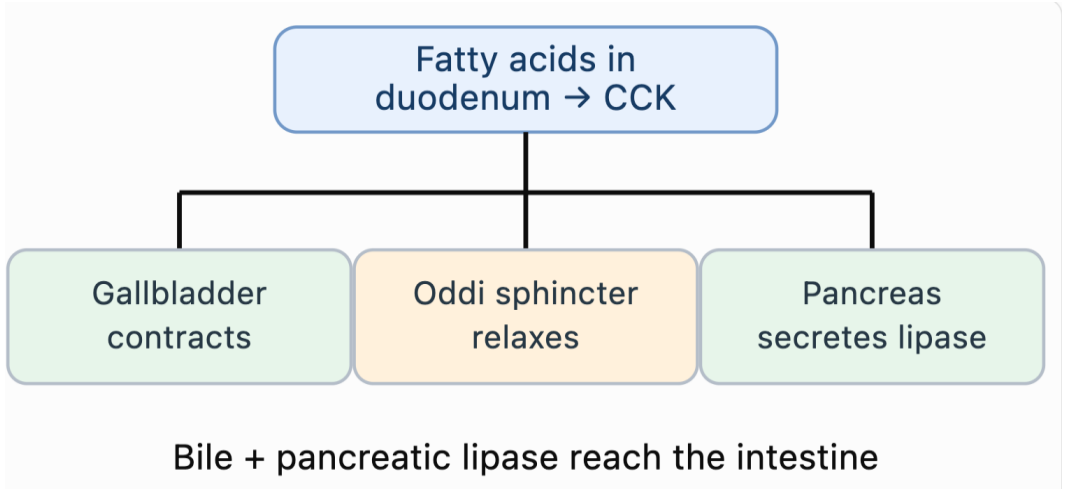
Cholecystokinin (CCK)

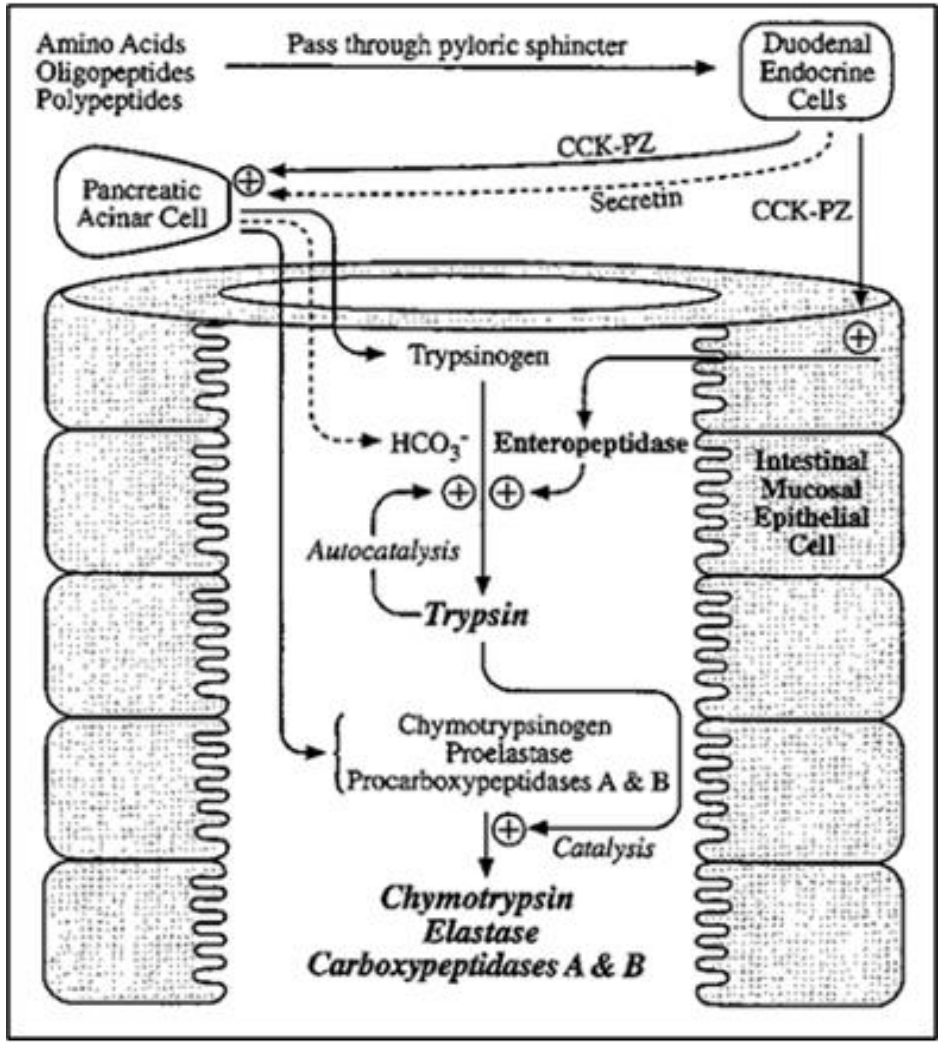
- Fatty acids in the duodenum stimulate CCK release.
- CCK stimulates:

Gallbladder contraction → Bile release

Relaxation of the sphincter of Oddi → Bile enters the duodenum

Pancreatic enzyme secretion → Pancreatic lipase



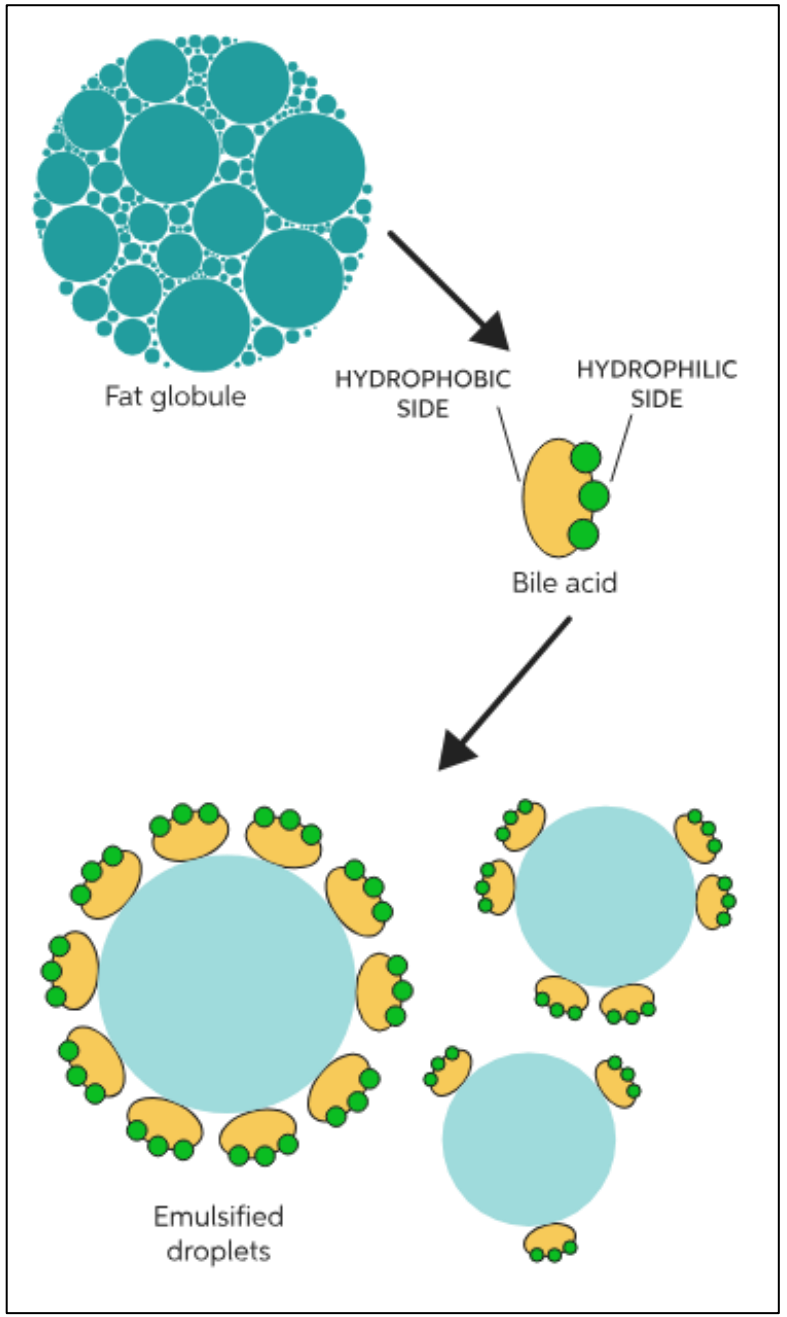


CCK coordinates bile delivery and pancreatic enzyme secretion.

Bile and Fat Emulsification

- Bile is produced by the liver and stored in the gallbladder.
- Bile salts help break large fat droplets into smaller droplets.
- This process is called emulsification.
- Smaller droplets provide a larger surface area for lipase.
- Pancreatic bicarbonate helps neutralise gastric acid.

Emulsification improves fat digestion.



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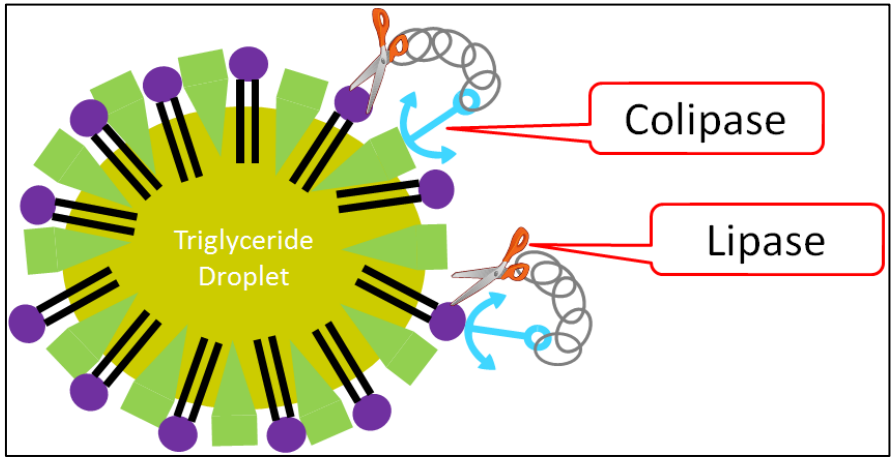
Pancreatic Lipase and Colipase

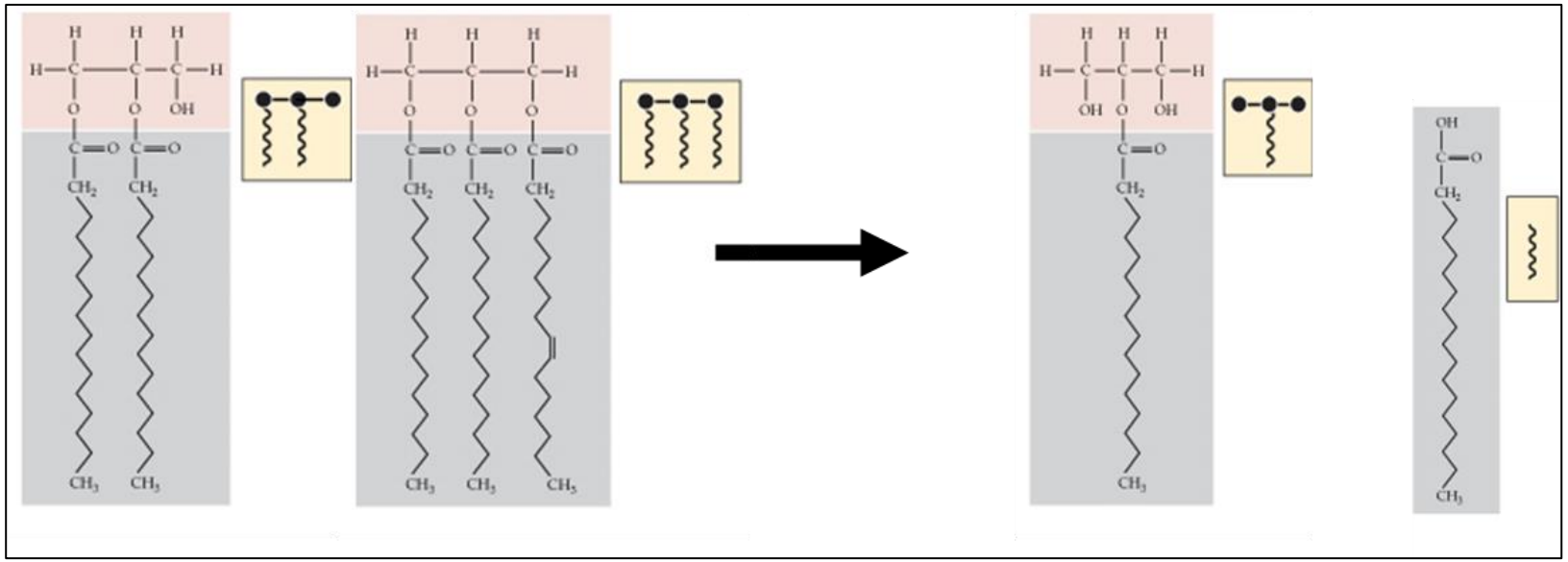
Pancreatic lipase

- Hydrolyses triglycerides at the surface of fat droplets.

Colipase

- Secreted by the pancreas.
- Helps lipase bind to fat droplets.
- Overcomes the inhibitory effect of bile salts on lipase binding.





Main products

Triglyceride → 2-Monoacylglycerol + 2 Fatty acids

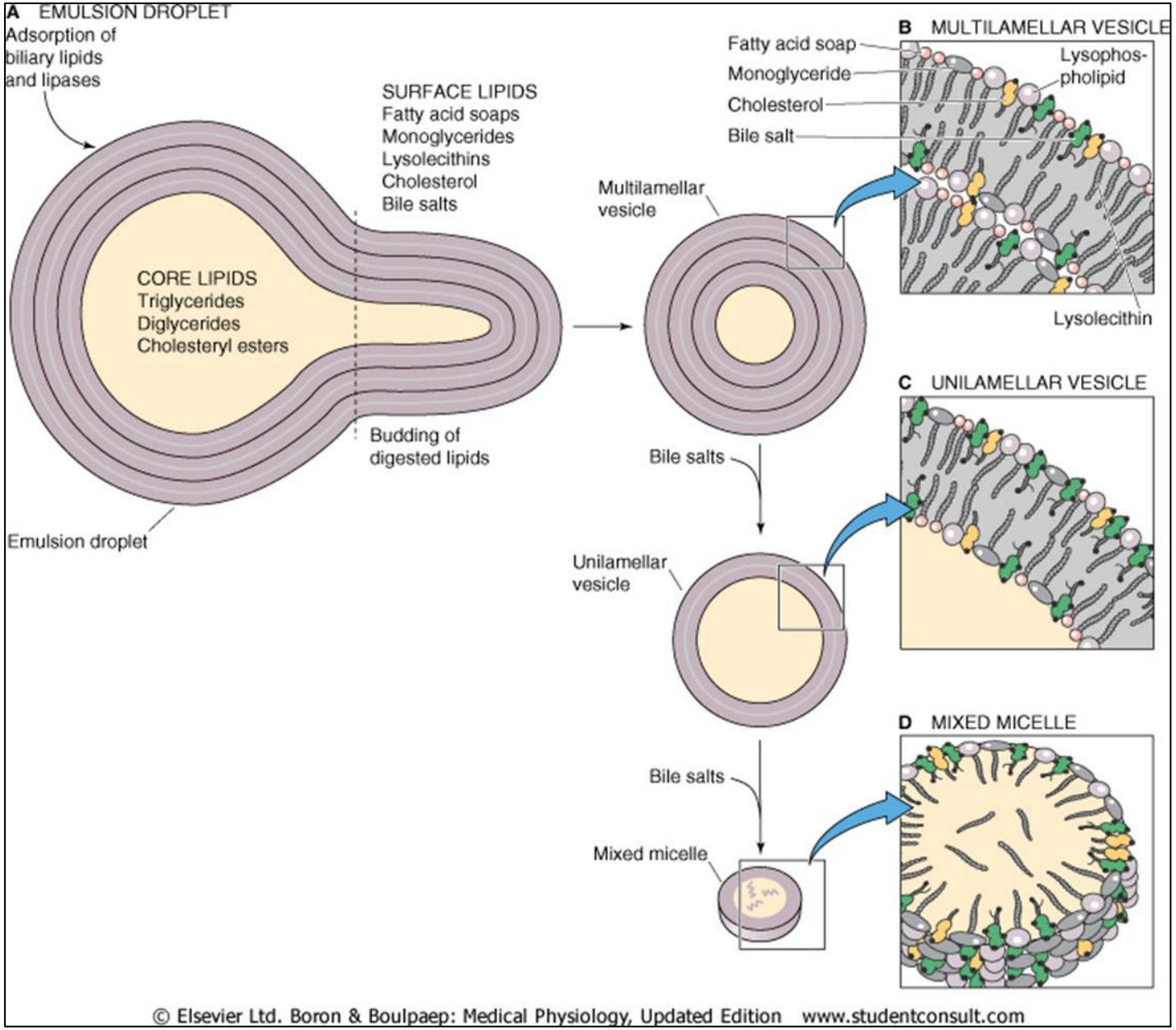
From Fat Droplets to Micelles

Formation of Mixed Micelles

1. Lipase hydrolyses triglycerides at the droplet surface.
2. Lipids from the core replace the digested surface lipids.
3. Fat droplets become smaller.
4. Lipid vesicles bud off.
5. Multilamellar vesicles become unilamellar vesicles.
6. Bile salts help form mixed micelles.

Micelles carry lipid digestion products towards enterocytes

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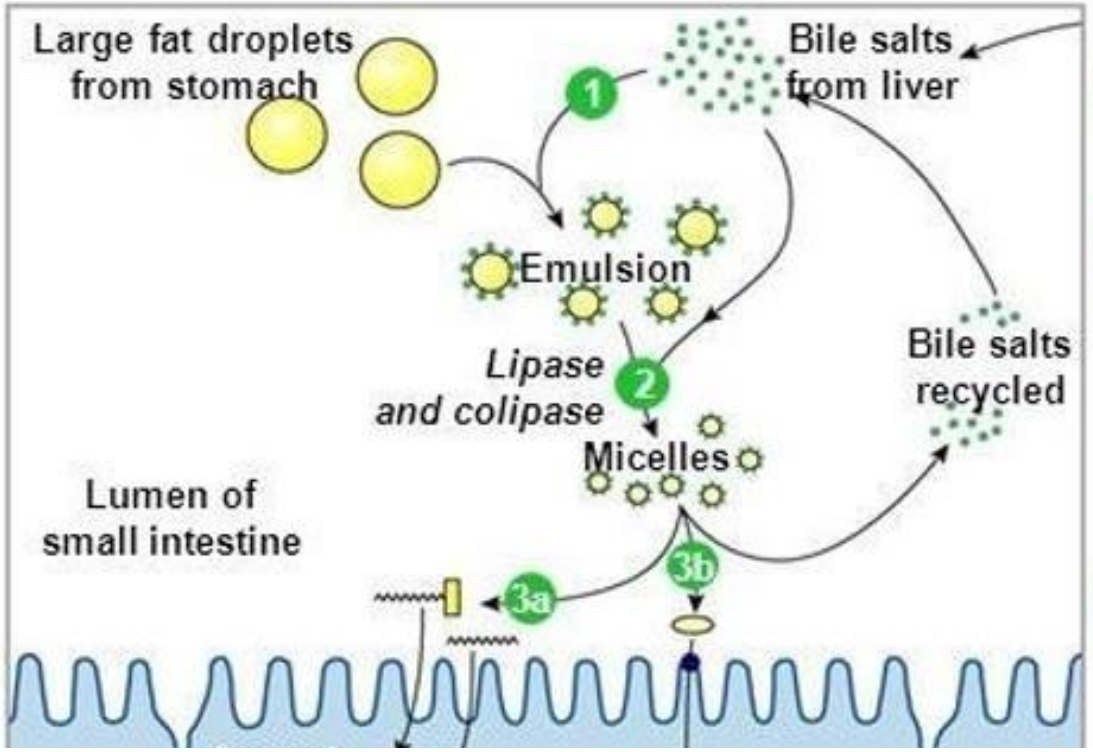
Digestion Summary: From Triglycerides to Micelles

Mouth and stomach

Lingual lipase + Gastric lipase → Partial digestion

Small intestine

- CCK → Bile release + Pancreatic lipase secretion
- Bile salts → Emulsification
- Pancreatic lipase + Colipase → Fatty acids + Monoglycerides
- Mixed micelles → Lipid absorption



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Fat Absorption and Transport

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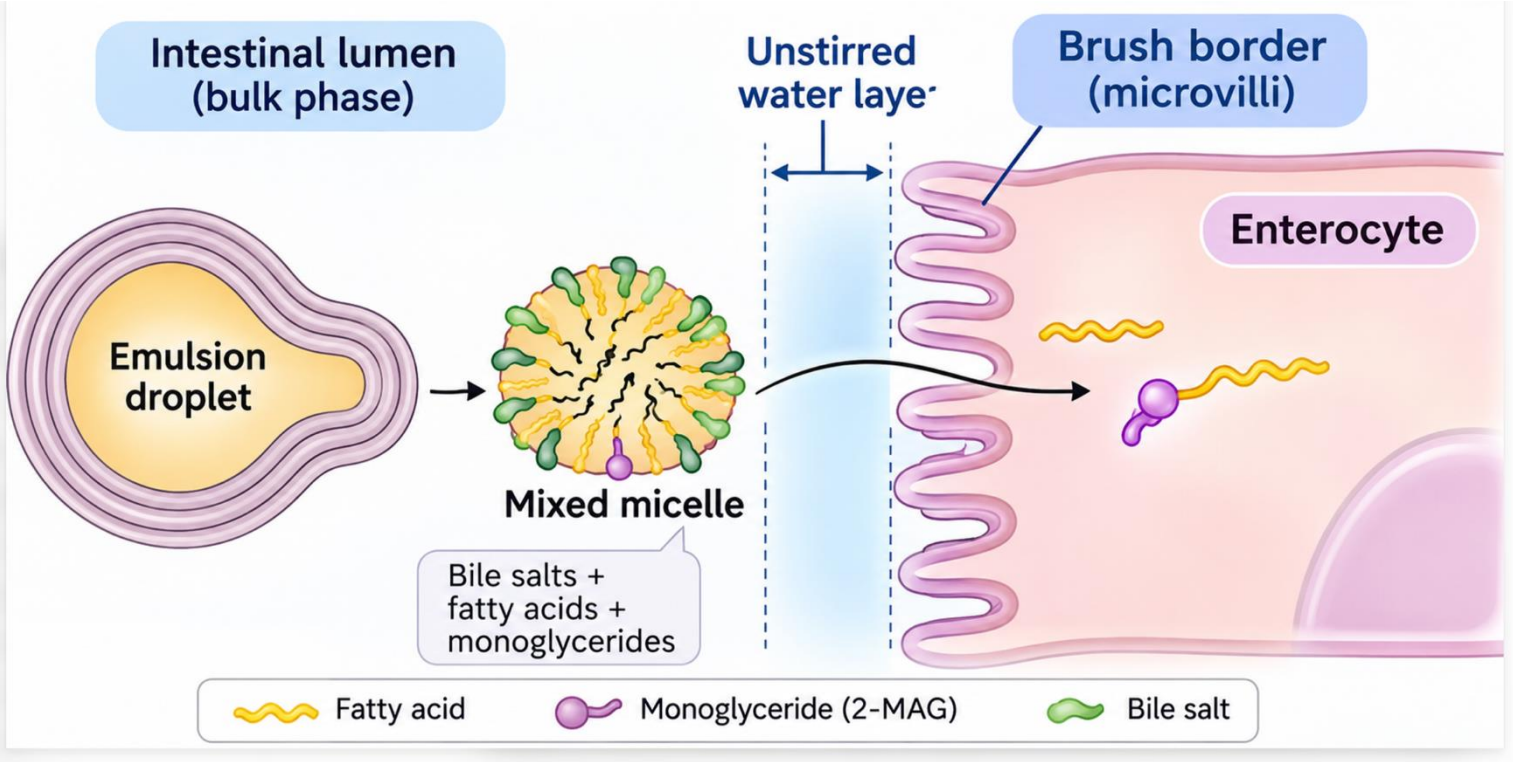
Absorption I: Entering the Enterocyte

Short- and medium-chain fatty acids

- More water-soluble.
- Can enter enterocytes without micelles (diffusions).
- Pass mainly into portal blood.

Long-chain fatty acids

- Poorly water-soluble.
- Mixed micelles carry them across the unstirred water layer.
- Enter enterocytes by diffusion, membrane incorporation or protein-mediated transport.



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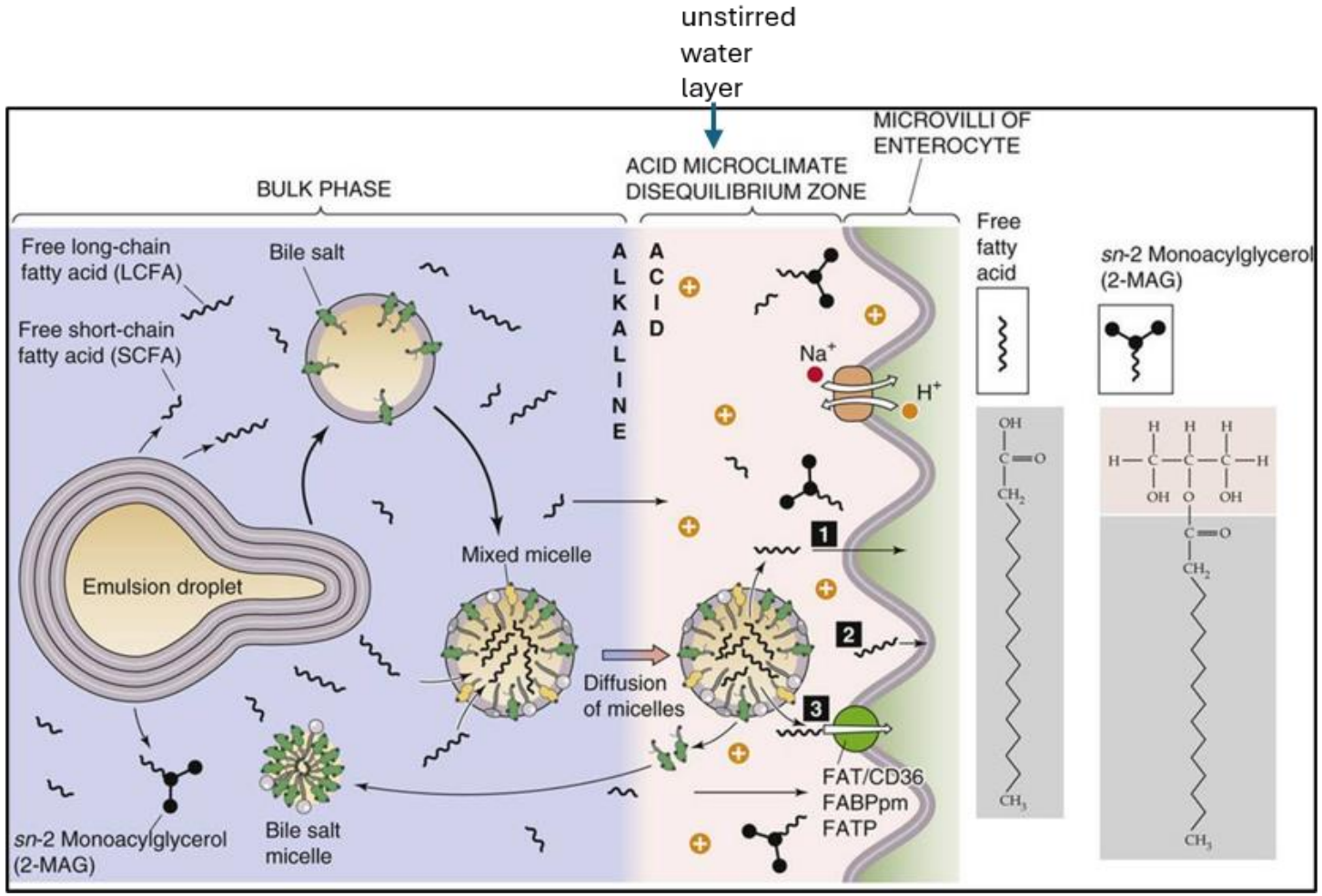
The Brush Border and Bile Salt Recycling

Brush-Border Absorption

- Na^+/H^+ exchange contributes to a slightly acidic microenvironment.
- Protonation can facilitate fatty acid uptake.
- Fatty acids enter by several mechanisms.

Bile salt recycling

- Bile salts remain mainly in the intestinal lumen.
- Reabsorbed primarily in the terminal ileum.
- Returned to the liver through portal blood.



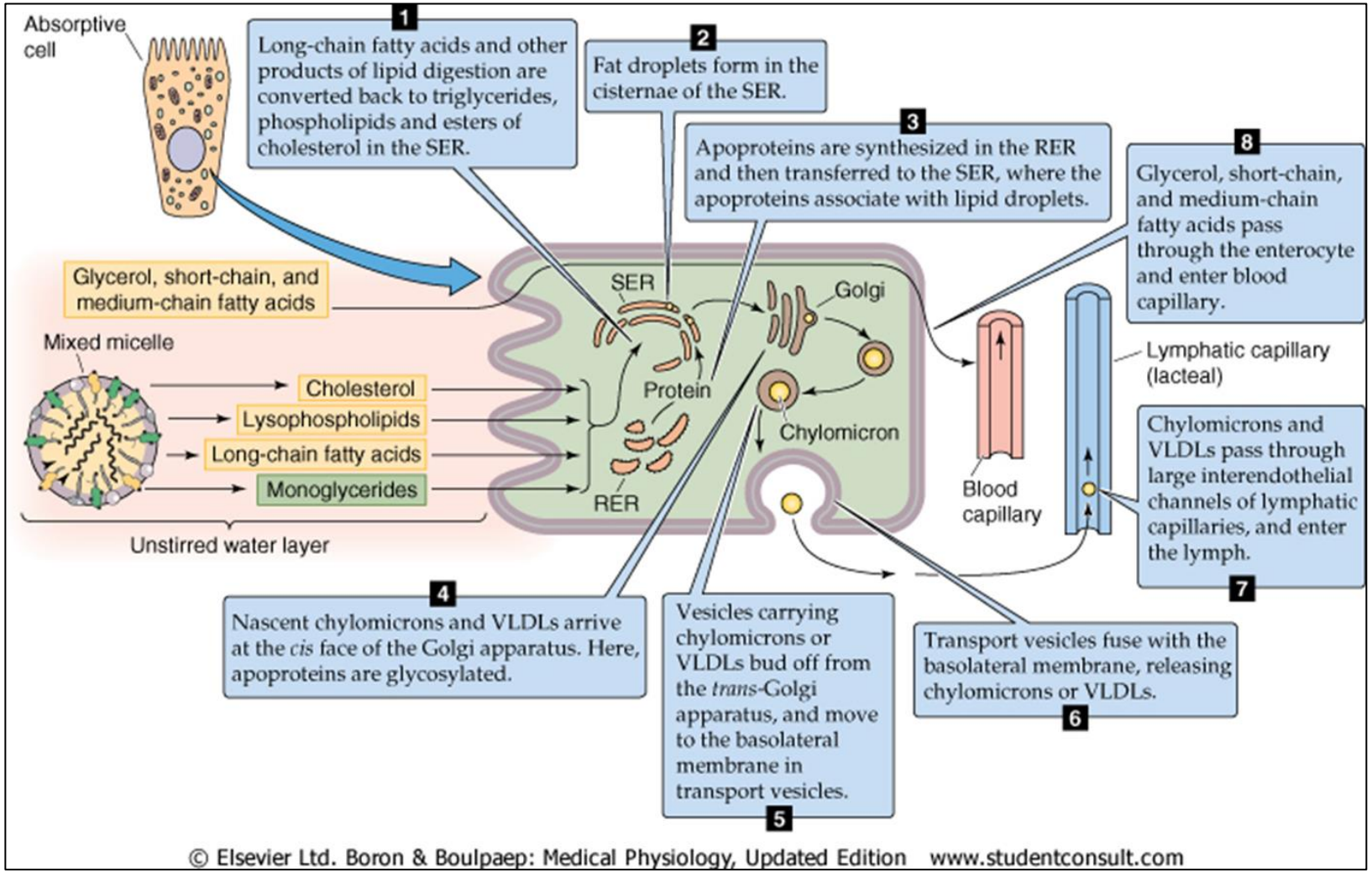
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Enterohepatic circulation = Recycling of bile salts

Absorption II: Lipid Processing Inside Enterocytes

Inside the Enterocyte

1. Long-chain fatty acids and monoglycerides enter the cell.
2. Triglycerides are re-formed in the smooth ER.
3. Lipids form droplets within the ER.
4. Apoproteins are synthesised in the rough ER.
5. Lipids and apoproteins form chylomicrons.
6. Chylomicrons are processed in the Golgi apparatus.
7. Vesicles release chylomicrons into lymphatic vessels.

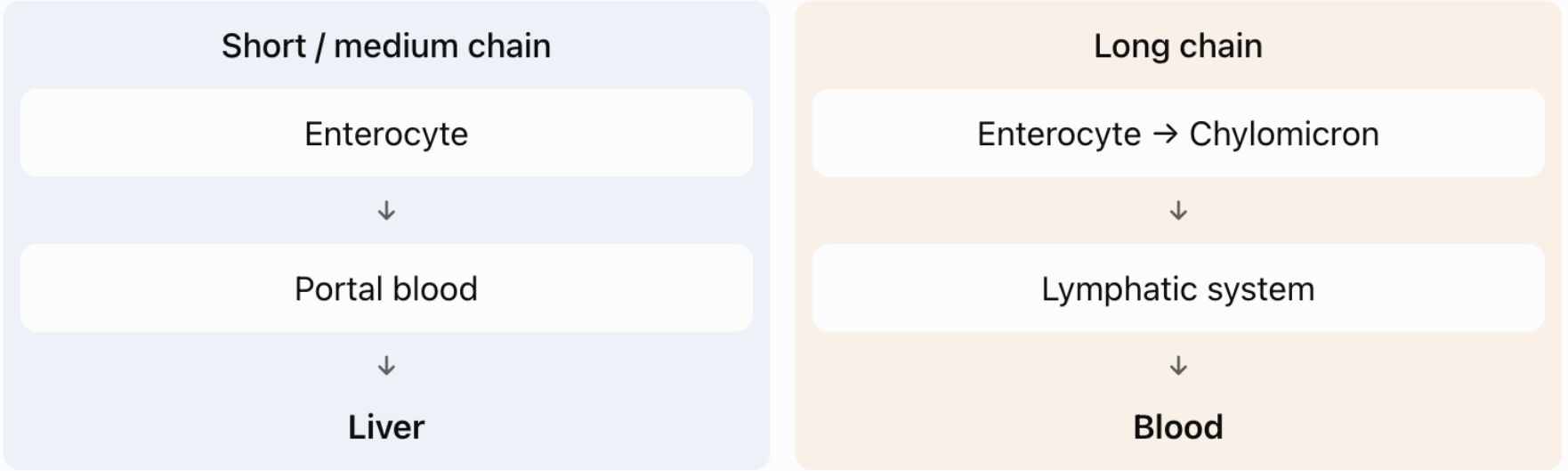


Main pathway: Enterocyte → Lymph → Blood

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Short-Chain vs Long-Chain Fatty Acid Absorption

Feature	Short/medium-chain FA	Long-chain FA
Water solubility	Higher	Lower
Micelles	Generally not required	Important
Inside enterocytes	Limited re-esterification	Re-formed into triglycerides
Main transport route	Portal blood	Lymphatic system
Transport form	Mainly free fatty acids	Chylomicrons

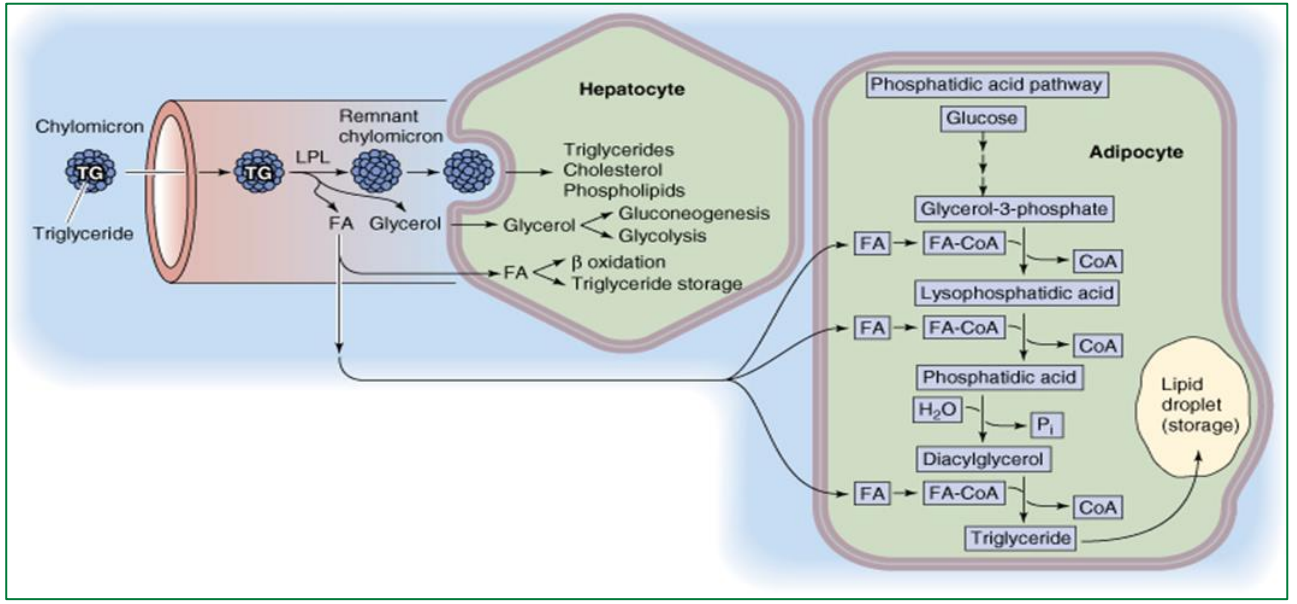


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Absorption III: Lipid Transport in the Blood

Chylomicrons in the Blood

- Chylomicrons transport dietary triglycerides through the blood.
- Lipoprotein lipase (LPL) acts at capillary endothelial surfaces.
- LPL releases fatty acids and glycerol from triglycerides.
- Fatty acids are taken up by muscle and adipose tissue.
- Chylomicron remnants are transported to the liver.
- Glycerol is also transported mainly to the liver



Fatty acids → Energy production or storage

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Chylomicron → LPL → Fatty acids → Muscle / Adipose tissue

Lipid Functions and Fat-Soluble Vitamins



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Lipids as an Energy Store

Functions of Lipids: Energy Storage

- Fat provides approximately 9 kcal/g of energy.
- Lipids are a major energy source during fasting.
- Non-polar lipids can be stored with very little water.
- Triglycerides are mainly stored in adipose tissue

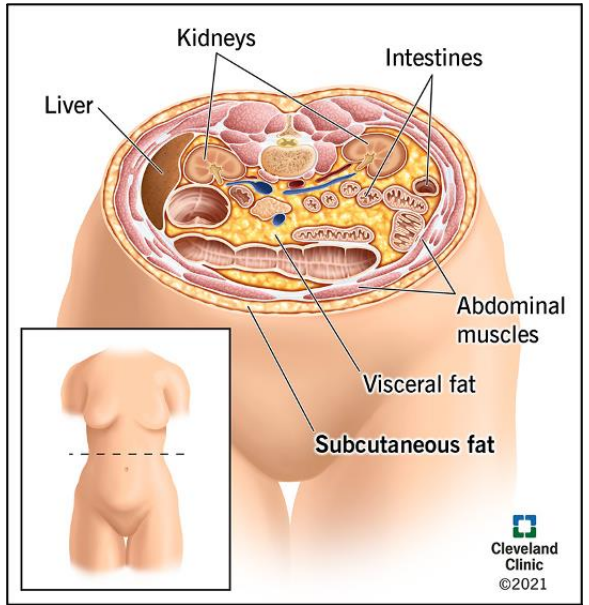
Stored Metabolic Fuel in a 70-kg Person

Constituent	Energy (kJ/g dry weight)	Dry Weight (g)	Available Energy (kJ)
Fat (adipose tissue)	37	15,000	555,000
Protein (muscle)	17	6,000	102,000
Glycogen (muscle)	16	120	1,920
Glycogen (liver)	16	70	1,120
Glucose (extracellular fluid)	16	20	320
Total			660,360

Sources: Owen, O. E., and Reichard, G. A., Jr., 1971. *Progress in Biochemistry and Pharmacology* 6:177; Newsholme, E. A., and Leech, A. R., 1983. *Biochemistry for the Medical Sciences*. New York: Wiley.

Fat storage locations

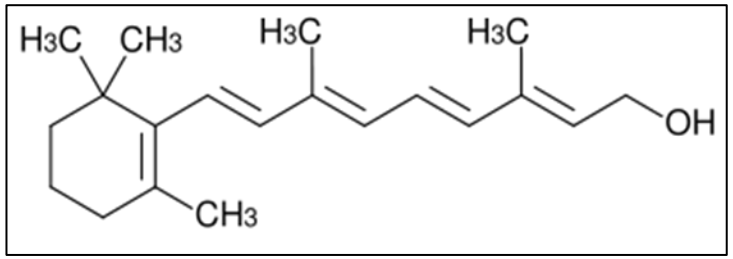
- Subcutaneous fat: beneath the skin.
- Visceral fat: around internal organs.
- Stored fat can provide energy for several weeks.



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Fat is an efficient long-term energy store.

Fat-Soluble Vitamins: Vitamin A

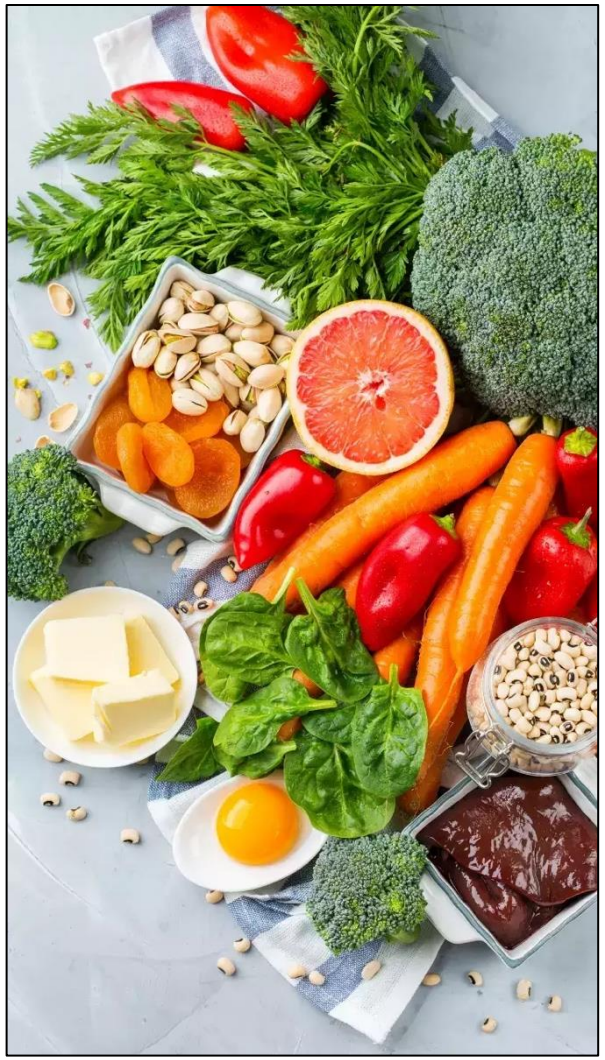


Fat-Soluble Vitamins: A, D, E and K

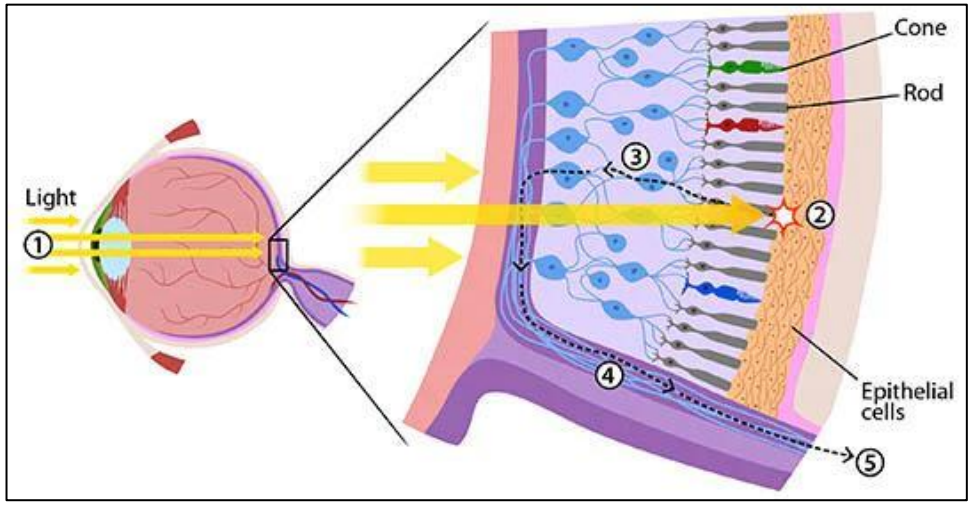
- These vitamins are absorbed with dietary lipids.
- Mixed micelles help deliver them to enterocytes.

Vitamin A

- Retinol: found in animal-derived foods.
- β-Carotene: found in plants; precursor of vitamin A.
- Essential for normal vision.
- Required for the visual pigment rhodopsin.



Deficiency → Night blindness



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Vitamin D: Bone Health

Sources

- Produced in human skin following sunlight exposure (vitamin D_3).
- Also obtained from dietary sources.
- Cats and dogs depend largely on dietary vitamin D.

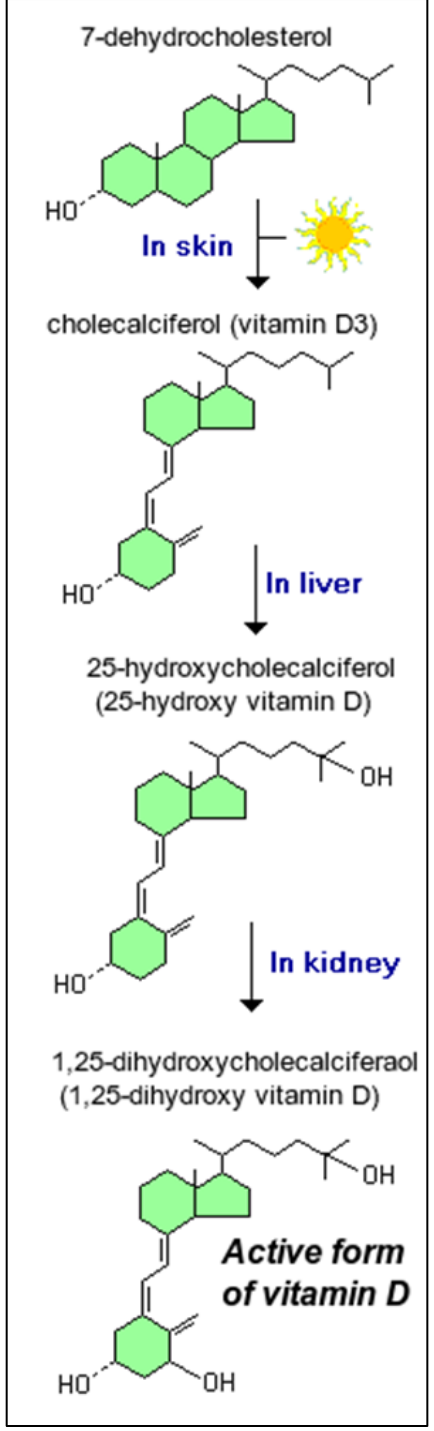
Activation

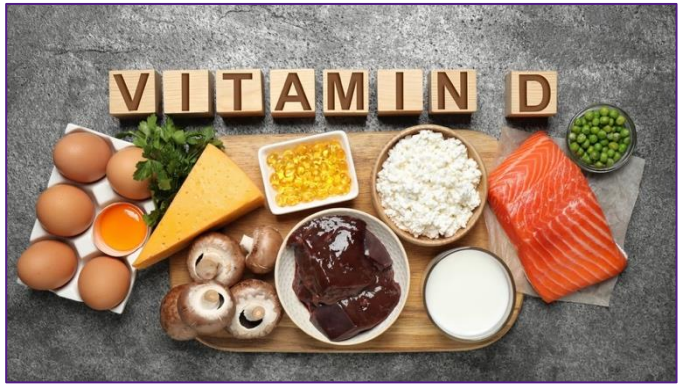
- Skin → Liver → Kidney
- Active form: Calcitriol

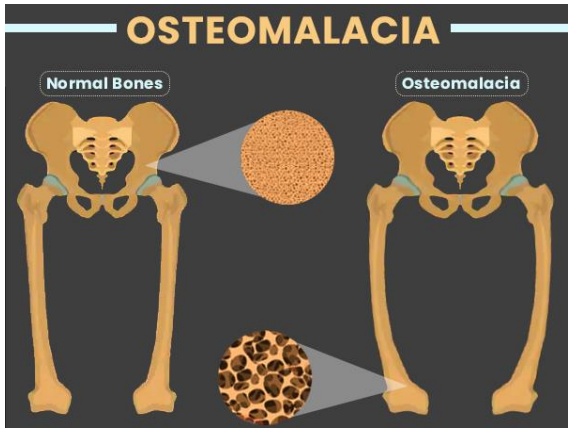
Functions and deficiency

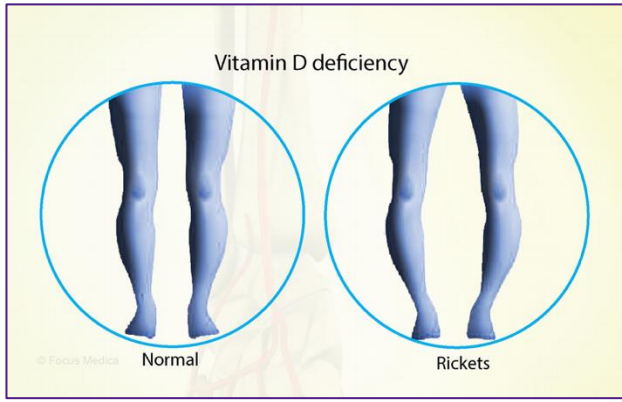
- Supports calcium and phosphate balance.
- Essential for normal bone mineralisation.
- Deficiency → Rickets (children) / Osteomalacia (adults).

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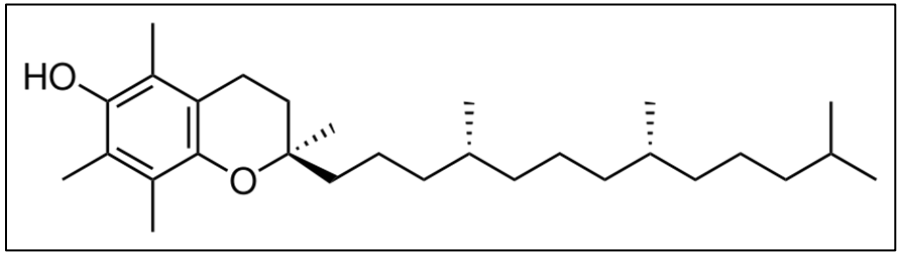


Vitamin E: Antioxidant Protection



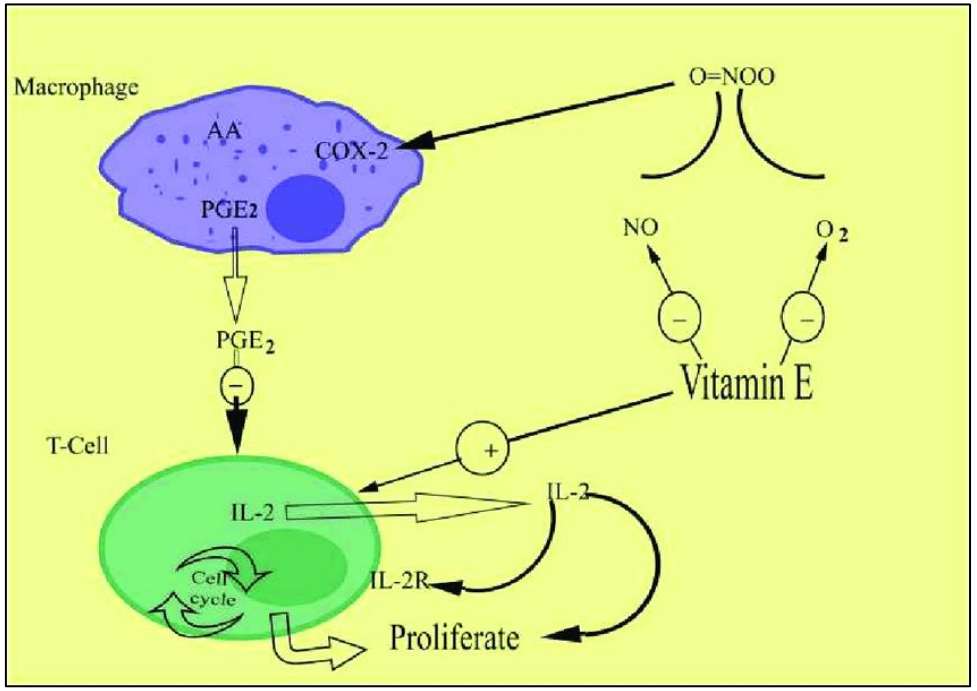
Vitamin E

- α-Tocopherol is an important form of vitamin E.
- Major dietary sources: vegetable oils, nuts and seeds.
- Acts as an antioxidant.
- Helps protect cell membranes from oxidative damage.



Reactive molecules

- Reactive oxygen intermediates (ROI)
- Reactive nitrogen intermediates (RNI)



Deficiency

- Rare.
- Reduced protection against oxidative damage.
- May cause neurological and neuromuscular problems.

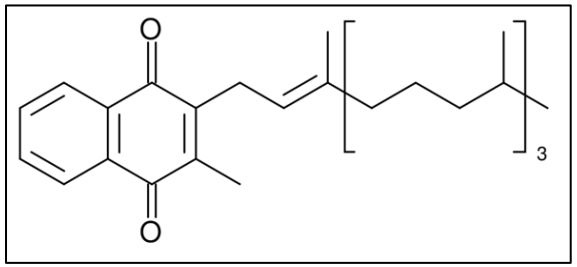
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Vitamin K: Blood Coagulation and Bone Health

Sources

- Green leafy vegetables.
- Some forms are produced by intestinal microbiota.





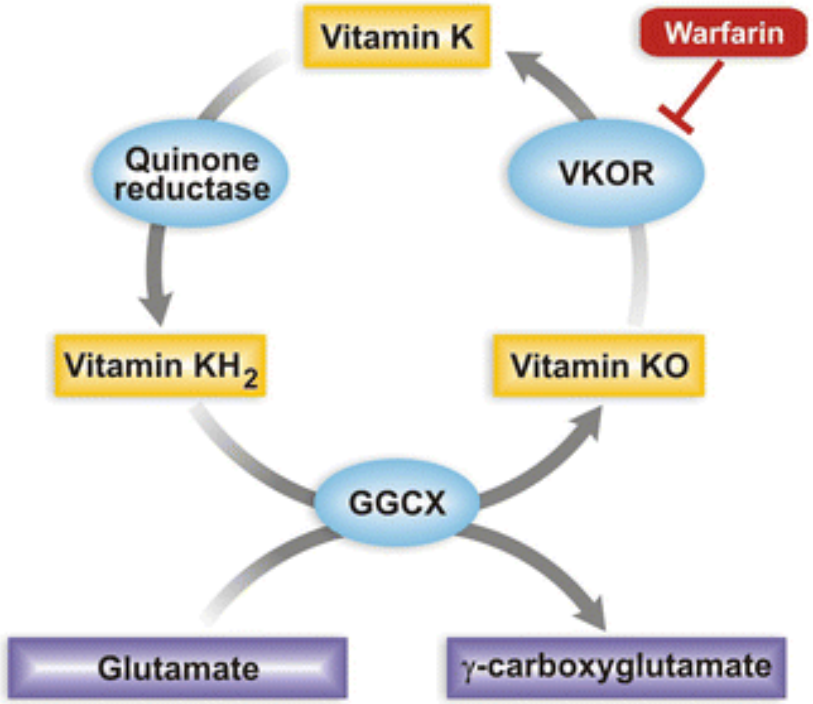
Mechanism

- Required for γ-carboxylation of specific glutamate residues.
- Produces γ-carboxyglutamate in vitamin K-dependent proteins.

Functions

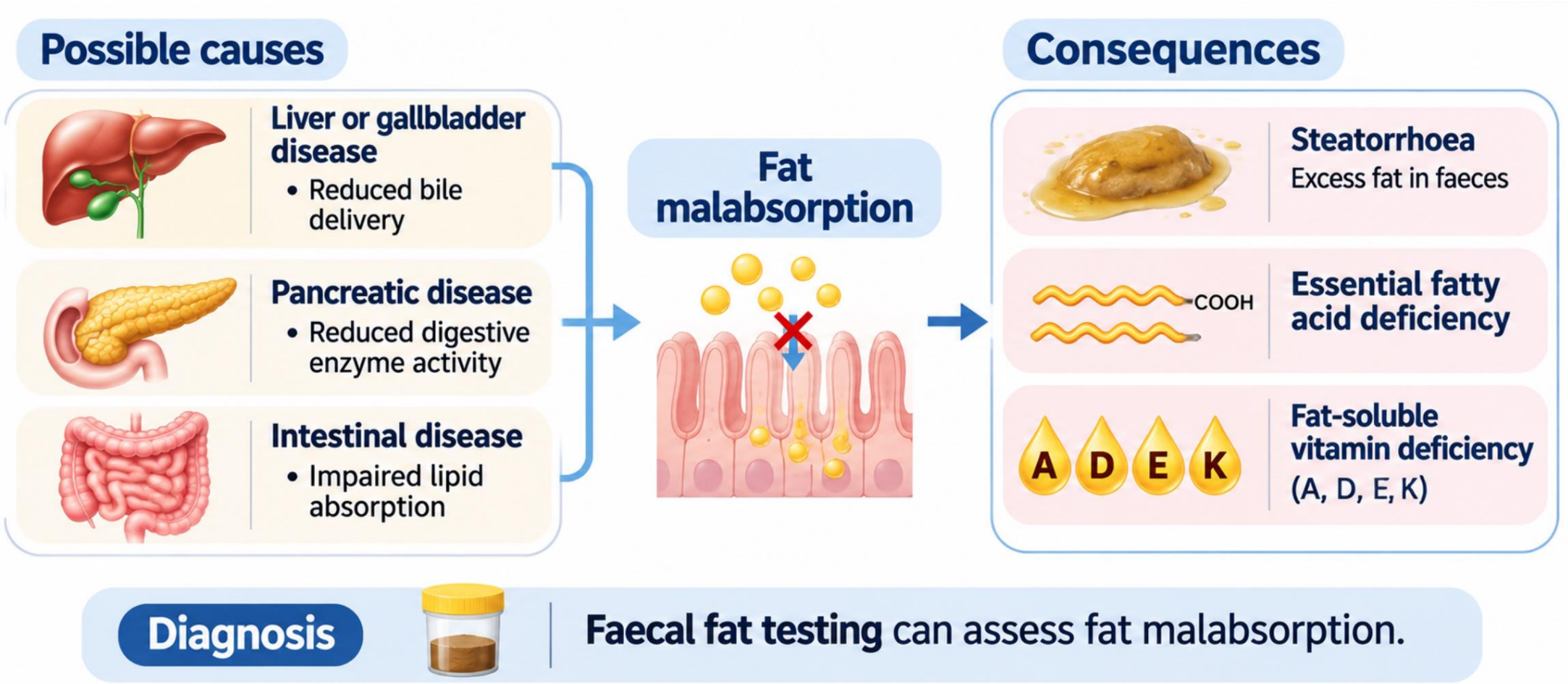
- Blood coagulation
- Bone metabolism
- Regulation of vascular calcification

Deficiency → Increased risk of bleeding



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Fat Malabsorption



Summary: Digestion and Absorption of Fats

1. Digestion

Lipases + Bile salts → Fatty acids + Monoglycerides

2. Absorption

- Short/medium-chain fatty acids → Portal blood
- Long-chain fatty acids → Chylomicrons → Lymph → Blood

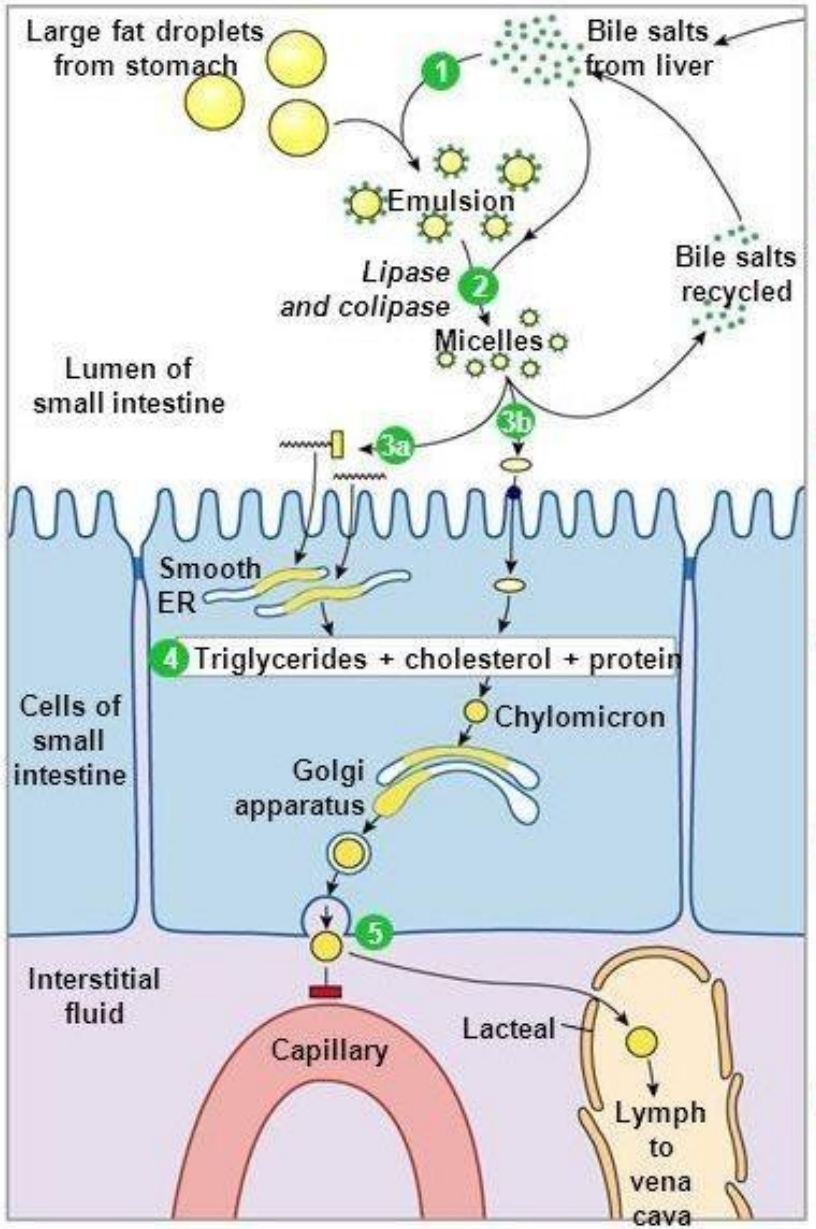
3. Functions

- Energy storage
- Cell membrane structure
- Essential fatty acids and signalling
- Absorption of vitamins A, D, E and K

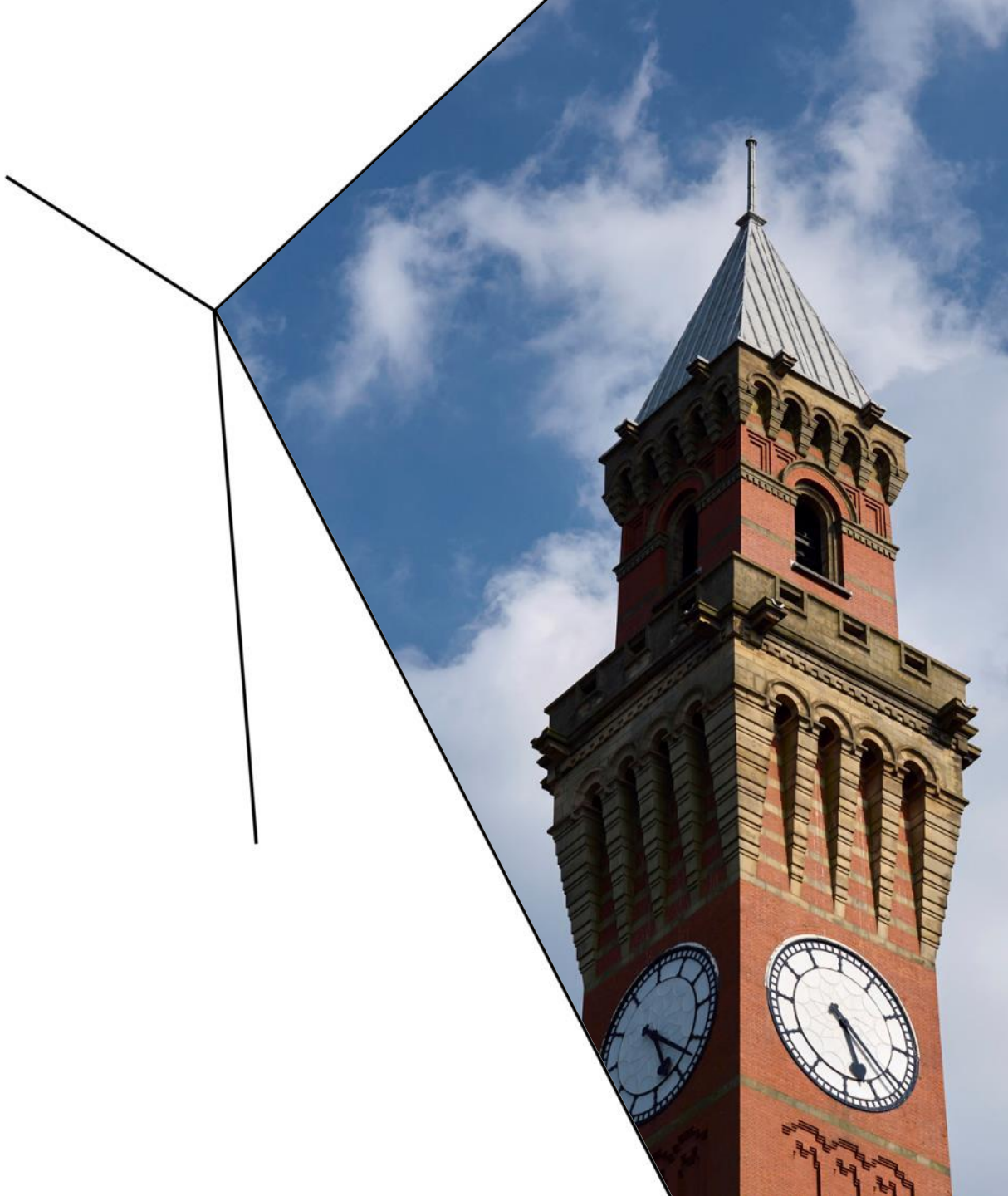
4. Clinical relevance

Fat malabsorption → Steatorrhoea + Nutrient deficiencies

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Questions



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